

Plan of Attack: Section 2, Sample Construction Plan

Costco Australia / Synthetic Control

Research question: Does the entry of a Costco gas station into an Australian local market cause nearby competitor stations to lower their retail fuel prices?

(a) Sample window

Endpoints. The pre-period for each treated Costco runs from the start of its state's price registry (August 2016 for NSW, January 2018 for WA, December 2018 for QLD) up to the day before that Costco's validated treatment date (see (d)). The post-period runs from the treatment date through April 2026, the most recent registry pull. Per-Costco pre-period lengths range from 26 (Perth Airport) to 58 months (Casuarina, Lake Macquarie); post-period lengths from 31 (Coomera) to 74 months (Perth Airport).

Pre-period length. Synthetic control fits its weights on the entire pre-period *trajectory* of the dependent variable, not just on a level. More pre-period observations tighten identification rather than dilute it, the opposite of the DID logic in which long pre-periods can introduce structural breaks into the parallel-trends assumption. Same-state donors experience the same world-shifts (oil-price regime changes, vehicle fleet turnover, federal excise adjustments) over the same window, so common shocks are absorbed into the synthetic by construction. We use as much pre-period as the data allow.

COVID. The 2020 demand collapse and recovery is the only structural break large enough to warrant explicit handling. Three of the four treated Costcos opened well after the COVID disruption (Lake Macquarie May 2022, Casuarina November 2022, Coomera May 2023); for those three the COVID period sits entirely inside the pre-period and is matched by the synthetic by construction. **Perth Airport (treated 19 February 2020)** opens approximately four weeks before Australia's COVID demand collapse begins; this is the most fragile case in the analysis. Its pre-period is entirely pre-COVID, but its early post-treatment months are dominated by pandemic-driven price movements rather than competitive response to Costco's entry. The synthetic counterfactual is built from WA donor postcodes that experienced the same pandemic shock, so the post-treatment gap nets out COVID provided the pre-period weights track the treated trajectory through January 2020. As a Section 3 robustness check, Perth Airport is re-estimated with March–December 2020 dropped from both panels.

(b) Other sample restrictions

Treated unit. For each treated Costco, the treated unit is the *local market* defined as all non-Costco retail stations within 5 km of the Costco's coordinates, summarised each month as the simple mean of those stations' monthly mean unleaded prices. We exclude Costco's own price quotes from the treated mean because the research question concerns competitor response, not Costco's own pricing.

Donor pool. Each donor unit is a 4-digit Australian postcode, summarised the same way (monthly mean of non-Costco unleaded price across all stations within the postcode). Postcodes are the natural geographic unit of the three state registries and are consistent across NSW, QLD, and WA. A postcode is excluded from the donor pool if any of its stations lie within 20 km of *any* Costco fuel station (treated, excluded, or otherwise), so that donor postcodes are unambiguously outside any plausible Costco competitive footprint.

Asymmetric radii (5 km treated, 20 km donor buffer). The IO literature on retail fuel-price competition (e.g. Houde 2012; Eckert 2013) finds price-spillover footprints concentrated within roughly 1–5 miles. 5 km (~3.1 mi) sits at the conservative end of that range, small enough to capture genuinely competing stations but not so small as to drop most of the local market. The 20 km donor buffer is intentionally a 4× multiple, providing a wide exclusion donut to ensure SUTVA holds: no donor postcode is partly ‘treated’ by an off-pool Costco. The asymmetry (small treated radius, large donor buffer) biases the design *against* finding an effect: narrow treated definition shrinks the treated population to the most exposed stations, and the wide buffer shrinks the donor pool to truly untreated postcodes.

Outcome variable. Regular unleaded (FuelCode in {U91, E10, P91} for NSW; Fuel_Type in {Unleaded 91, e10} for QLD; PRODUCT_DESCRIPTION = ULP for WA). Unleaded is the highest-volume retail product in Australia, comparable across the three state schemes, and Costco’s headline competitive product.

Donor-pool measurement-quality filters. A postcode is additionally excluded if it averages fewer than 3 unique stations per month (monthly means based on 1–2 stations are too noisy to serve as credible counterfactual building blocks) or has fewer than 24 months of any observed coverage (weights cannot be reliably fit on a short pre-period).

Treated-Costco inclusion rule. A Costco fuel station is treated as a synthetic-control unit if and only if its registry history contains both **at least 24 pre-treatment months** and **at least 6 post-treatment months** in the relevant state’s price registry. Four of Australia’s ten Costco fuel stations satisfy both conditions (Coomera, Casuarina, Perth Airport, Lake Macquarie); six do not, and appear as descriptive context in Section 1’s coverage table. Exclusion reasons: Marsden Park (11 pre-months), Auburn (8 post-months), Canberra Airport (no pre; ACT joined NSW FuelCheck after this Costco opened), Casula and North Lakes (≤4 pre-months each; the Costco predates the relevant registry), Ipswich (3 pre-months, predates the QLD scheme).

Table 1. Threshold-and-defense summary.

Threshold	Value	Justification	Robustness check (Section 3)
Treated radius	≤5 km	Conservative end of IO-literature retail fuel-price spillover range (1–5 mi)	Re-run at 3 km and 8 km
Donor buffer	>20 km	4× the treated radius; ensures SUTVA, asymmetric to bias against effect	Re-run at 15 km and 30 km
Donor stations / month	≥3	Below 3, monthly mean has high sampling variance	Re-run at ≥2 and ≥5
Donor coverage	≥24 months	Minimum pre-period length for stable SC weight fitting	Re-run at ≥36 months
Costco pre-period	≥24 months	Same logic, applied to treated	Not relaxed (would re-introduce excluded Costcos that fail the rule)
Costco post-period	≥6 months	Minimum to estimate a sustained-effect gap	Not relaxed
Plausible price range	50–500 ¢/L	Sentinel-value floor and ceiling; AU retail unleaded lives in 80–280	None; not an analytic threshold

(c) Dependent variable integrity

Construction. The dependent variable is the monthly mean unleaded retail price (cents per litre) at the treated 5 km ring or the donor postcode, computed in two stages: first within each (station $\times$ month), then across stations in the ring or postcode. Section 1's Figure 1 shows the post-aggregation distribution ($n = 17,953$ postcode-months, range 92–259 ¢/L, approximately bimodal across the pre-2022 / post-2022 price-regime divide).

Sentinel-value filter. Individual price quotes outside 50–500 ¢/L are dropped before aggregation. This is a data-integrity filter, not winsorisation: it removes obvious registry artifacts (out-of-stock placeholder values like 9999 or 0.01) rather than legitimate tail observations. Australian retail unleaded post-2018 has averaged 130–220 ¢/L; anything outside 50–500 is mechanical, not market.

Missing months. A postcode-month or treated ring-month with zero contributing station observations is left as missing (NaN), never imputed as zero, never carried forward. Synthetic-control fitting handles missing months by ignoring them in the pre-period objective; Section 1 reported that the largest run of missing months in any treated panel is three (Lake Macquarie), which is not material.

No winsorisation in the main specification. Two-stage monthly aggregation smooths individual-quote outliers: an extreme single quote at one station gets averaged with the rest of that station's month, and then with the rest of the ring or postcode. If a robustness check exposes a donor-postcode month driving an estimate, we winsorise the analysis panel at the 99th percentile and report both specifications side-by-side.

(d) Independent variable integrity

Treatment indicator. A binary post-opening flag is constructed per treated Costco by merging the hand-collected treatment-date table onto the (Costco $\times$ month) panel: 1 from the treatment month forward, 0 before.

Treatment-date validation. Registry first-appearance can lag the true fuel-station opening if a station begins selling fuel before appearing in the state's reporting feed. Each of the four treated Costcos' registry-first-appearance dates was cross-checked against Costco Australia corporate announcements, trade press, and news coverage. The corroborating source must specifically reference the *fuel station* rather than the warehouse; warehouse opening dates are not used because fuel stations and warehouses at Costco Australia regularly open weeks apart. If a fuel-station-specific announced opening date differs from registry first-appearance by more than 30 days, the announced date supersedes.

Costco	Registry first-appearance	Fuel-station-specific corroboration	Δ	Treatment date used
Coomera	23 May 2023	23 May 2023 (Retail World; explicit fuel-station opening)	0 d	23 May 2023 (registry)
Casuarina	11 Nov 2022	None; trade press only reports the warehouse opening (7 Dec 2022)	n/a	11 Nov 2022 (registry)
Perth Airport	11 Apr 2020	19 Feb 2020 (ACAPMAg: petrol station selling at \$1.18/L; warehouse opens 19 Mar 2020)	–52 d	19 Feb 2020 (announced)

Lake Macquarie	2 May 2022	None; no fuel-station-specific opening date in public sources	n/a	2 May 2022 (registry)
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Three of the four registry first-appearance dates either match the gas-station-specific corroborated date exactly (Coomera) or lack any gas-station-specific external date for comparison (Casuarina, Lake Macquarie); these retain the registry date. Perth Airport is the only revision: the ACAPMAG report explicitly distinguishes the petrol station (open 19 Feb 2020, with bowser price quoted) from the warehouse (opening one month later on 19 March 2020), placing the fuel-station opening 52 days before the first FuelWatch record.

Station coordinates and distance-to-Costco. Distance is computed via the haversine great-circle formula from each station's coordinates to the nearest Costco fuel station's coordinates. NSW and WA historical registry files do not carry lat/lng, so coordinates are joined from one-time pulls of the live FuelCheck and FuelWatch APIs by (normalized name, postcode); QLD's historical files include coordinates inline. Stations that fail the coordinate join are dropped from analysis (they cannot be classified into a 5 km ring or a donor postcode without a known location); the per-registry join failure rate is reported in (f).

(e) Aggregation

Why monthly. Three considerations point at month as the right level of aggregation. First, NSW and QLD are event-based registries (rows are price-change events) while WA is a daily-snapshot registry; month is the lowest common frequency at which all three speak the same language without having to invent 'daily' observations for the event-based feeds. Second, the research question is about sustained competitive response to Costco entry, which plays out over weeks-to-months rather than days, so daily resolution would add noise without identification. Third, monthly means across multiple stations in a ring or postcode smooth single-quote outliers without requiring any winsorisation step.

Two-stage aggregation. Within each station $\times$ month, we compute a single mean unleaded price across all of that station's price observations in the month. We then take a simple (unweighted) mean across stations to produce the (ring $\times$ month) treated panel and the (postcode $\times$ month) donor panel. The two-stage structure prevents stations that change prices more often (and therefore appear more times in the event-based NSW/QLD feeds) from contributing disproportionately to the cell mean.

Carry-forward step (NSW and QLD only). For NSW and QLD, where rows are price-change events rather than daily snapshots, we construct each station's daily price series by carrying its last-observed price forward to subsequent days within the month until the next observed change. This is the standard approach for converting price-event data to a daily series and then to monthly means. WA's daily-snapshot data passes through unchanged.

(f) Data merging

Three sources are joined into a single analysis panel:

- **State registries** (NSW FuelCheck, QLD FPRS, WA FuelWatch) provide the price observations.
- **Live-API station snapshots** (NSW, WA) and inline coordinates (QLD) provide station latitude / longitude, joined to historical registry rows by (normalised station name, postcode).

- **The hand-collected Costco metadata** (`data/catalogs/usable_costcos.csv`) provides each Costco's lat/long and treatment date, joined to treated panels by Costco name.

Expected losses in the merge. Two sources of leakage. First, stations that closed, rebranded, or were re-numbered before the April 2026 live-API pull are absent from that snapshot, so historical rows that mention them fail the coordinate join. The snapshot misses **2,376 of 4,543 NSW historical stations (52.3%)** and **673 of 1,423 WA historical stations (47.3%)**; QLD is unaffected because its historical files carry coordinates inline. Row-level impact after the upstream fuel-type and sentinel-price filters is reported in (g). Second, station-name typos and mid-panel renames occasionally break the join. We address renames programmatically where we can (Section 1 noted the May 2022 NSW Costco ‘Members only’ suffix; the build script strips it). The loss falls almost entirely on the donor pool (all four treated Costcos join cleanly), and 196 donor postcodes survive the downstream geographic and quality filters.

(g) Filters and transformations: the funnel from raw to analysis-ready

Counts are exact at every step. Row units in the upper block give way to (station × month) cells, (postcode × month) cells, postcode counts, and Costco counts as the panel is aggregated and filtered. Source: `scripts/synthetic_control_input/build_sc_inputs.py`.

Table 2. Sample-construction funnel.

Step	NSW	QLD	WA	Justification
Raw registry pull (rows)	4,401,971	4,970,650	8,047,153	Source data
Filter to regular unleaded (rows)	1,982,507	997,680	2,059,755	Question is about retail unleaded competition; premium/diesel are different markets
Drop sentinel prices (50–500 ¢/L; rows)	1,982,506	995,145	2,059,755	Removes out-of-stock placeholder values; nearly nonbinding once the unleaded and date filters have run
Join station coordinates (rows)	1,247,728	995,145	1,289,912	Required for radius assignment; NSW/WA join from live-API snapshots (per-registry failure rates in (f)); QLD has inline coords
Aggregate to (station × month)	156,864	70,564	68,332	Carry-forward for NSW/QLD events; passthrough for WA snapshots
Aggregate to (postcode × month)	42,307	18,186	18,089	Donor candidate panel (561 / 245 / 198 postcodes) before geographic and quality filters
Drop postcodes ≤20 km from any Costco	–231 pcds	–51 pcds	–80 pcds	SUTVA: donors must be unambiguously outside Costco's competitive footprint
Drop postcodes with <3 stations / month avg	–226 pcds	–138 pcds	–82 pcds	Donor monthly mean too noisy below 3 stations
Drop postcodes with <24 months observed	–0 pcds	–0 pcds	–0 pcds	Coverage was good; no postcode failed this gate

Final donor pool	104 pcds	56 pcds	36 pcds	196 postcodes × ~90 months = 17,578 (postcode × month) cells
Costco eligibility (≥24 pre, ≥6 post)	−4 Costcos	−2 Costcos	0	Inclusion rule applied to all 10 Australian Costco fuel stations (5 NSW, 3 QLD, 2 WA)
Final treated panel	1 Costco	1 Costco	2 Costcos	4 Costcos × (5 km ring) × ~80–100 months = 375 (Costco × month) cells

The output is four files in `data/sc_inputs/`: `treated_units.csv` (375 Costco-month rows), `donor_pool.csv` (17,578 postcode-month rows), and two metadata files.